

# Trader Performance vs. Bitcoin Market Sentiment

Analysis of Hyperliquid Historical Trade Data vs. the Fear & Greed Index

## 1. Objective

This analysis explores the relationship between trader performance on Hyperliquid and prevailing Bitcoin market sentiment, as measured by the daily Fear & Greed Index (FGI). The goal is to identify patterns in win rate, profitability, trading volume, and positioning bias across sentiment regimes (Extreme Fear, Fear, Neutral, Greed, Extreme Greed), and to translate these patterns into concrete, actionable trading-strategy recommendations.

## 2. Datasets & Data Cleaning

**Historical Trader Data** (Hyperliquid): 211,224 trade records across 32 accounts and 246 traded assets, spanning May 2023 to May 2025. Fields used: Account, Coin, Execution Price, Size (Tokens/USD), Side, Direction, Closed PnL, Fee, and Timestamp.

**Bitcoin Fear & Greed Index:** 2,644 daily sentiment readings (Feb 2018 – May 2025), classified into five buckets: Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

### Cleaning steps performed:

- Parsed the *Timestamp IST* field (DD-MM-YYYY HH:MM) into proper datetimes and derived a day-level *date* key.
- Verified the raw trade dataset had zero missing values across all 16 columns.
- Left-joined trades to the daily sentiment classification on *date*; only 6 of 211,224 trades (<0.01%) had no matching sentiment day and were dropped.
- Separated trades into all trades (for volume/positioning analysis) vs. PnL-realizing ("closed") trades — those with non-zero Closed PnL — for win-rate and profitability analysis, since many rows represent position-opening trades with no PnL yet.
- **Data caveat:** the assignment brief references a *leverage* field; the actual data export provided does not include one, so leverage effects could not be analyzed and are excluded rather than estimated.

## 3. Summary Metrics by Sentiment

| Sentiment     | Trades | Volume (USD) | Win Rate | Avg PnL/Trade | Total PnL (USD) |
|---------------|--------|--------------|----------|---------------|-----------------|
| Extreme Fear  | 21,400 | \$114.5M     | 76.2%    | \$71.0        | \$739K          |
| Fear          | 61,837 | \$483.3M     | 87.3%    | \$112.6       | \$3.36M         |
| Neutral       | 37,686 | \$180.2M     | 82.4%    | \$71.2        | \$1.29M         |
| Greed         | 50,303 | \$288.6M     | 76.9%    | \$85.4        | \$2.15M         |
| Extreme Greed | 39,992 | \$124.5M     | 89.2%    | \$130.2       | \$2.72M         |

## 4. Key Visualizations

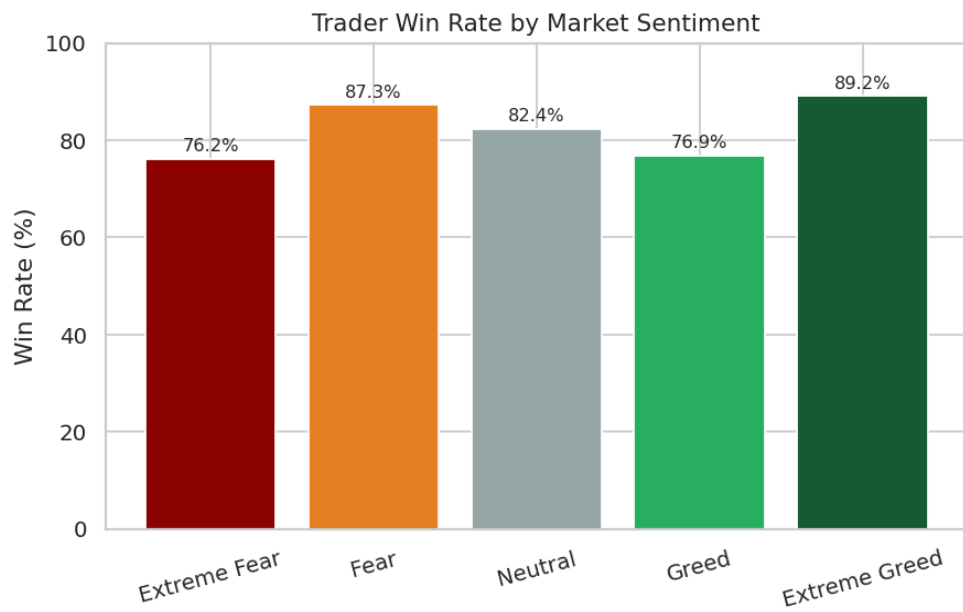


Figure 1. Win rate is highest in Extreme Greed and Fear, lowest in Extreme Fear and plain Greed — a non-linear pattern.

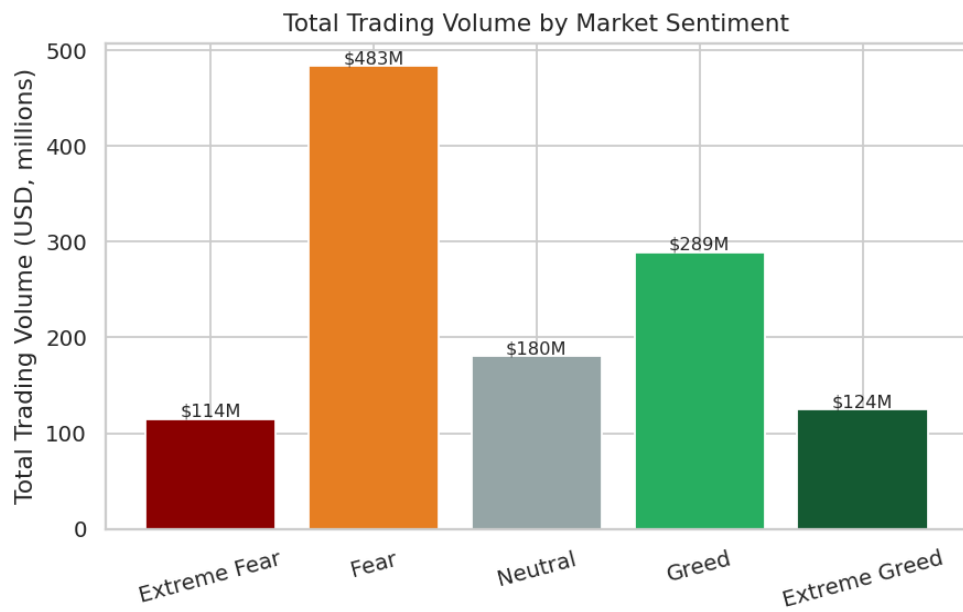


Figure 2. Trading volume peaks during Fear, not Greed — traders are most active in fearful markets.

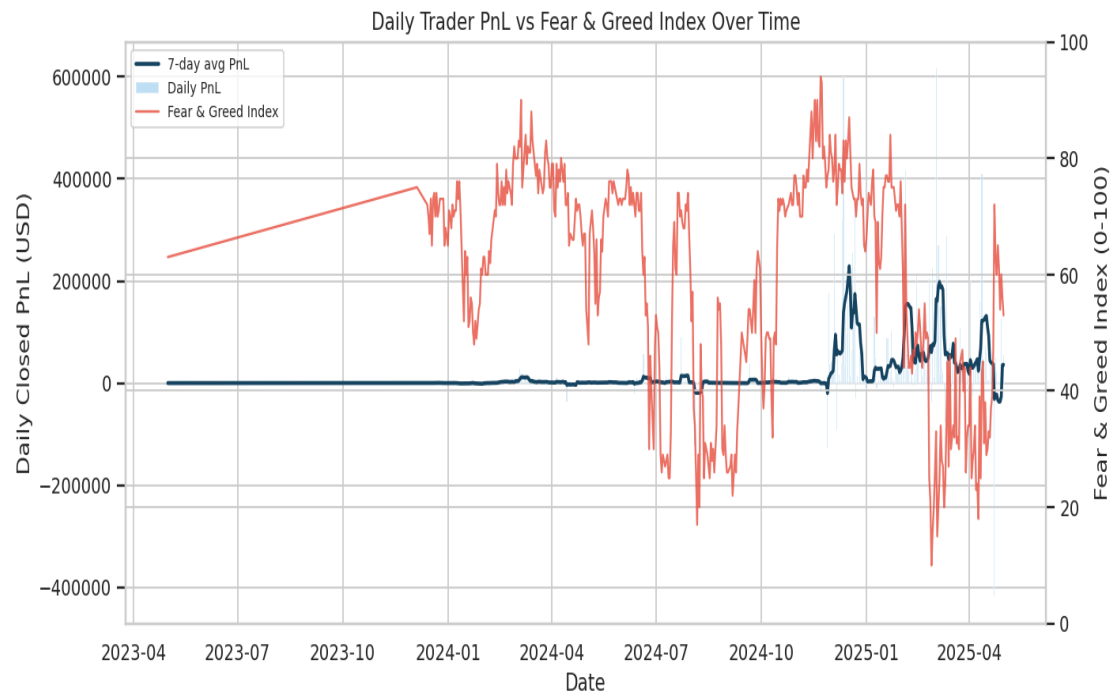


Figure 3. Daily realized PnL (bars, 7-day average in blue) vs. the Fear & Greed Index (red) over time.

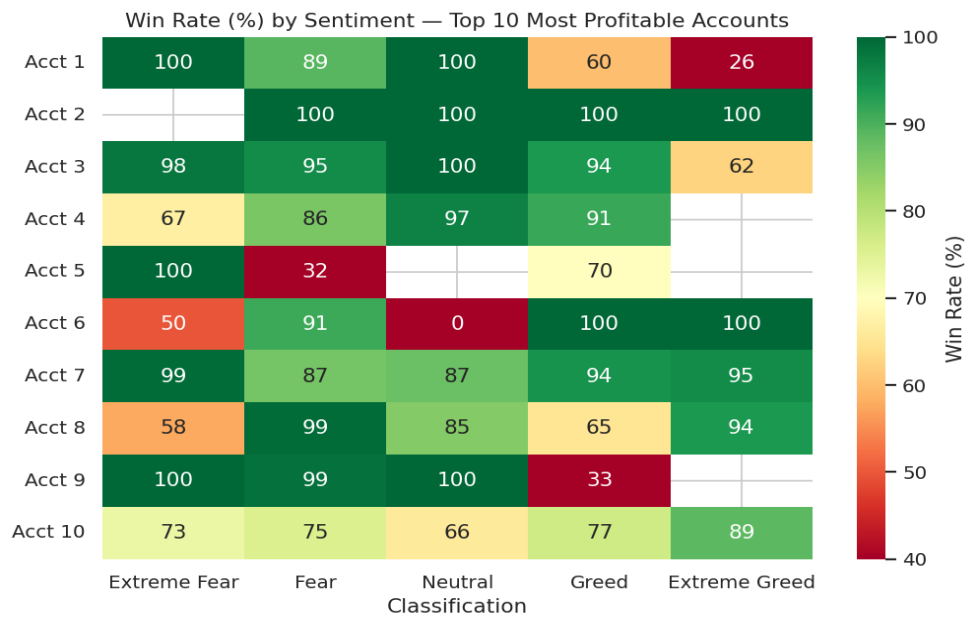


Figure 4. Win rate by sentiment for the 10 most profitable accounts — elite traders stay consistently sharp across every regime.

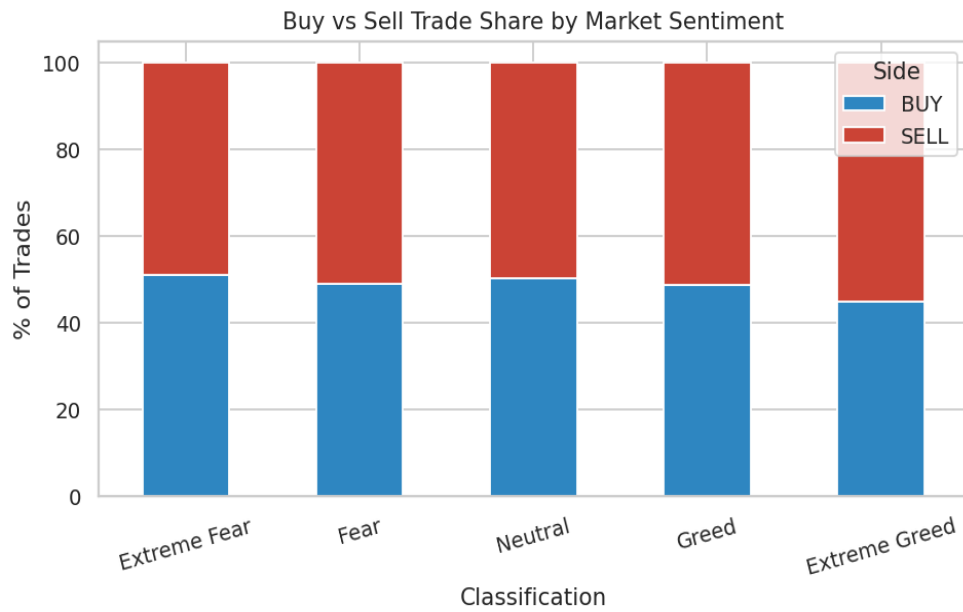


Figure 5. Sell-side share of trades rises from Extreme Fear (49%) to Extreme Greed (55%), consistent with profit-taking as euphoria builds.

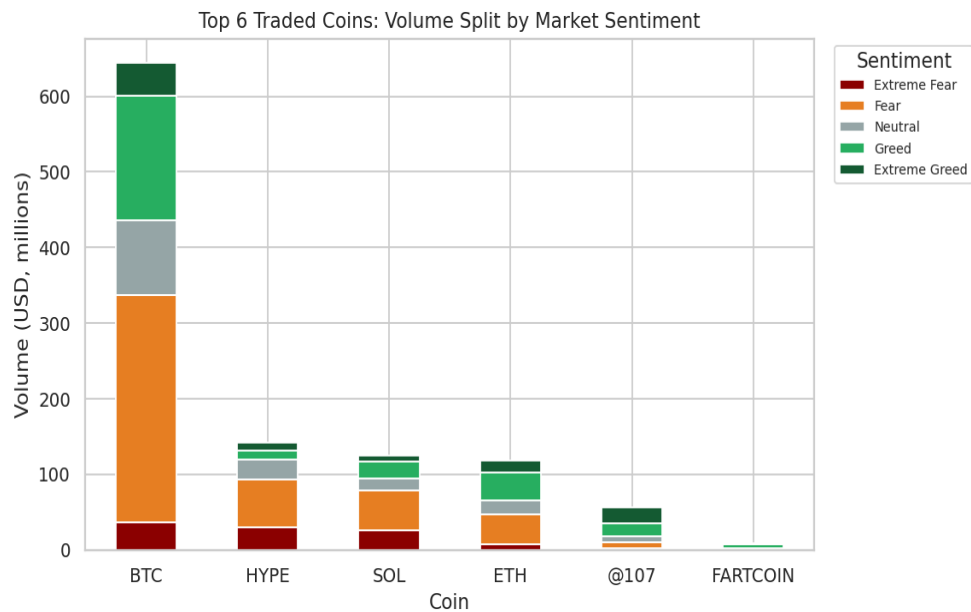


Figure 6. BTC, HYPE, SOL and ETH dominate volume in every regime; alt/meme-coin activity (e.g. FARTCOIN) grows in Greed phases.

## 5. Key Insights

- **Win rate is not monotonic with greed.** Extreme Greed (89.2%) and Fear (87.3%) show the highest win rates, while Extreme Fear (76.2%) and plain Greed (76.9%) are weakest. The riskiest periods are moderate-greed euphoria-building and outright panic/capitulation — not the sentiment extremes themselves.
- **Fear drives the most volume and the most total profit.** Fear accounted for the highest trading volume (\$483M) and the highest total realized PnL (\$3.36M) of any sentiment bucket — traders in this sample are disproportionately active, and successful, when markets are fearful.
- **Sell-side skew rises with greed** (49% of trades in Extreme Fear vs. 55% in Extreme Greed), consistent with traders fading strength and taking profit as euphoria builds.
- **Performance is highly concentrated.** The top 2 accounts alone generated over \$3.7M in combined PnL, while several accounts in the sample were net negative overall.
- **Elite traders are largely sentiment-agnostic.** The 10 most profitable accounts sustain high win rates (mostly 70–100%) across all five sentiment regimes — suggesting a durable trading edge rather than simple exposure to market mood.
- **Major coins dominate across all regimes** (BTC, HYPE, SOL, ETH), with a modest tilt toward speculative/meme assets during Greed and Extreme Greed.

## 6. Recommendations for Trading Strategy

- **Lean into Fear, cautiously.** Historical win rates and volume reward liquidity providers/dip buyers who transact during Fear; consider scaling entries during Fear/Extreme Fear rather than waiting for confirmed Greed.
- **De-risk during moderate Greed.** The lowest win rate outside Extreme Fear occurs in plain Greed — a transitional zone with uncertain trend continuation. Use tighter stop-losses and smaller position sizes here.
- **Copy elite-trader discipline, not sentiment timing.** Since top performers maintain edge across all regimes, prioritize consistent risk management (sizing, exits) over trying to time sentiment swings.
- **Watch rising sell-side share as a top-signal.** An increasing SELL/BUY ratio as sentiment approaches Extreme Greed is a useful real-time cue for anticipating local tops.
- **Size speculative/alt-coin exposure conservatively during Greed phases,** where their volume share rises but average win rates are comparatively lower.

## 7. Limitations

The dataset does not include a leverage field despite being referenced in the assignment brief, so leverage-adjusted risk could not be assessed. Sentiment is measured daily while trades occur intraday, introducing some resolution noise. The sample covers only 32 accounts, so conclusions about "trader behavior" broadly should be validated on a larger account set before informing capital allocation decisions.